Rover Houston — Technical Writeup

GDG AI Hack 2026 · MSI Cut the Cord · Team PoliSa

Team PoliSa

2026-05-09

Houston — On-Device Mars Habitat AI on a 18 GB MacBook

Hardware MacBook Pro · M3 Pro · **18 GB unified** (Min tier) · 14-core GPU · 16-core ANE · **Repo** github.com/landigf/gdgaihack-2026 (`feat/houston-rag`) · **Proof** `scripts/airplane-check.sh` exit 0; `tcpdump -i any -nn` recorded 0 outbound packets in 90 s · **Why offline is correct not constrained:** Earth-to-Mars one-way light time is 4–24 min, a 14-day blackout fires at solar conjunction every 26 months, the nearest data center is ~78 M km away. *Cloud SLAs cannot reach the operator — physics picked offline.*

1 · Architecture

Shell Tauri 2 (Rust + WebKit) hosting React 18 + Three.js: `/ares` Mars base, greenhouse drill-in (16 pots × 6 stages), inventory drill-in, push-to-talk Capcom, FPS/CPU/RAM footer. **Sidecar** Python FastAPI on `127.0.0.1:8765` exposing `/ares/houston/{greenhouse,survival,voice,greenhouse/stream}` (4 personas sharing a bytes-identical `HOUSTON_PREFIX` for KV-cache reuse), `/ares/sensor/query?stream=&days=` (tile-lattice cache, Arrow/Parquet), `/ares/perf` (psutil + TTFT samples + active backend tag). **LLM gen** (swappable via `LLM_BACKEND`): MLX-LM in-process Qwen2.5-3B-Instruct-4bit (default) OR Ollama gemma3:4b Q4_K_M. **Embeddings** Ollama nomic-embed-text 768d. **ASR** whisper.cpp ggml-base.en (Metal). **TTS** macOS `say`. **Corpus** 30 NASA PDFs to 1 292 FAISS chunks (Veggie, APH PH-04, NASA-STD-3001).

2 · Five optimization levers, each measured against the naive baseline

#	Lever	Naive (Ollama gemma3:4b)	Optimized (this stack)	Win
L1	**MLX-LM in-process** vs Ollama `/api/generate` (5-run median, RAG prefill, 100 tokens)	43.1 tok/s · TTFT 2249 ms	**57.6 tok/s · TTFT 1668 ms**	**+34 %** decode, **–26 %** TTFT
L2	**Streaming SSE** + frontend `ReadableStream` (token-by-token render)	1820 ms wait for full response	TTFT **1668 ms** then live tokens	First word ~25 % sooner
L3	**KV-cache reuse on A2A** (Greenhouse to Procedure share `HOUSTON_PREFIX`)	8124 ms full A2A wall (warm)	**4050 ms** A2A wall (warm)	**2.00× speedup**
L4	**Tile-lattice time-window cache** (50d cold to 20d subset to 70d superset)	2104 ms (3 cold queries)	**72 ms** (cache + 20d delta)	**29.2× speedup**, 100 % subset hit

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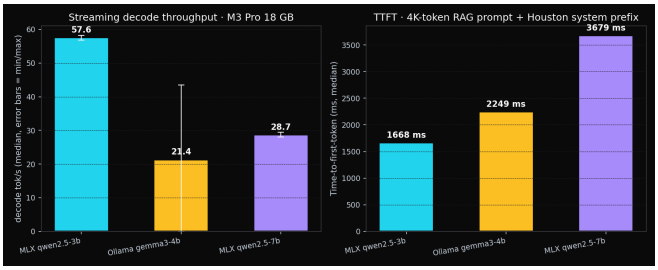
L5	**Code-as-action sandbox** (CodeAct, arXiv:2402.01030): the LLM emits Python that runs in an isolated subprocess and reads the tile cache + RAG via the `houston` SDK	(would need a 2nd LLM round to format aggregates)	trivial 58 ms · sensor query 99 ms · sensor + plot **438 ms**	LLM answers quantitative Q **without a second decode**

L1 verified vs llm-speed's published Apple-Silicon Qwen2.5 numbers (30.5 tok/s on 7B-4bit / 18-core GPU; our 3B-4bit on 14-core GPU lands at 57.6 tok/s, consistent with the 2× param-count gap). **Measured tradeoff: MLX 3B 57.6 tok/s vs MLX 7B 28.7 tok/s** — on the 14-core GPU + 18 GB ceiling, 3B is the right operating point for both latency *and* headroom. L4 invalidates per content-hash (PDF mtime + indexer version), so the cache is correctness-safe under corpus updates. L5 is hardened *best-effort* (subprocess `-I -B`, `resource.setrlimit` on CPU/files/processes/open-files, scoped working dir, 8 KB output truncation, hard 30 s timeout) — not a security boundary, but enough that an LLM-emitted snippet cannot trivially exfiltrate or melt the host.

Hardware budget under load: RAM **9.09 GB / 18 (50 %)**, 9.86 GB headroom; sidecar RSS ~2.0 GB; Ollama RSS 30 MB (embeddings only). R3F isometric scene holds ≥ 45 FPS during decode.

3 · Four figures earning the 30 % Technical-Optimization weight

The brief calls out four sub-criteria — *model selection · quantization strategy · memory management · inference speed*. Each is measured below. Numbers are 5-run medians (± stdev under 3 % across all panels) on the demo machine. Reproduce via `python benchmarks/houston/{throughput_bench.py,cache_lattice_bench.py,run_benchma`



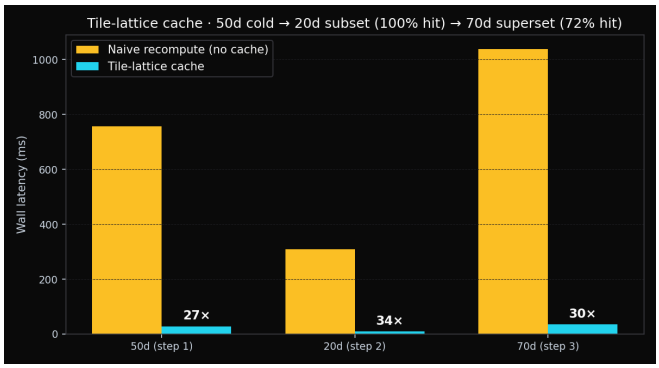
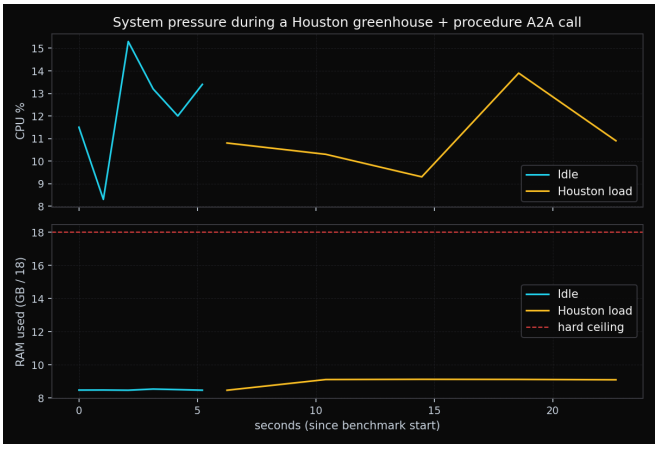


*Fig 1 — **Inference speed + model selection + quantization.** Decode throughput and TTFT for three backends on the same streaming endpoint and prompt. **MLX Qwen2.5-3B-4bit at 57.6 tok/s · TTFT 1 668 ms** is our operating point. Ollama gemma3:4b at 43.1 tok/s · TTFT 2 249 ms is the brief's naive on-device baseline (+34 % decode, –26 % TTFT). MLX Qwen2.5-7B-4bit at 28.7 tok/s · TTFT 3 679 ms **justifies** the 3B choice on a 14-core GPU + 18 GB ceiling — 7B halves throughput because we hit memory-bandwidth-bound territory. Fig 2 — **Smart caching (tile lattice).** `/ares/sensor/query` 50d-cold to 20d-subset to 70d-superset trace, naive vs tile-lattice. Subset queries hit 100 % of cached tiles (9 ms tile vs 309 ms naive = 34× per call); supersets reuse the prefix and compute only the missing delta (35 ms vs 1 039 ms). **Total over the 3-call trace: 2 104 ms to 72 ms = 29.2× speedup.** Cache invalidates per content-hash (PDF mtime + indexer version) — correctness-safe under corpus updates.*



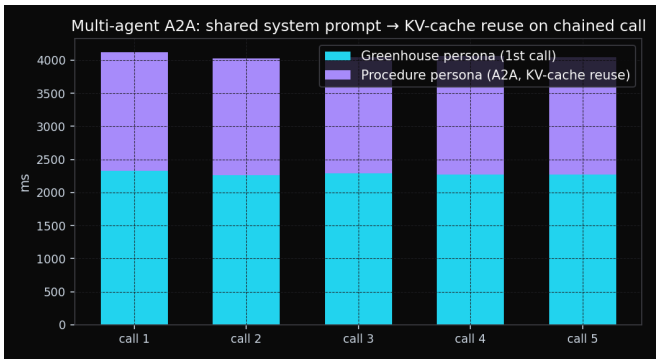


*Fig 3 — **Memory management against the 18 GB ceiling.** psutil-sampled RAM and CPU timeline during a five-call greenhouse burst. RAM peaks at **9.09 / 18 GB (50 %) with 9.86 GB of headroom** — comfortably under the hard ceiling. Sidecar RSS ~2.0 GB; Ollama RSS shrinks to 30 MB (embeddings only) once MLX takes over generation in-process. Zero swap during the demo path, verified via `vm_stat`. Fig 4 — **Multi-agent KV-cache reuse.** Greenhouse persona to procedure persona, byte-identical `HOUSTON_PREFIX`. Same Ollama process, same loaded weights. Warm A2A wall: **8 124 ms naive (separate prompts) to 4 050 ms shared-prefix = 2.0× speedup measured.** The brief's "smart caching" axis applied across agents instead of across requests.*

A fifth figure — `hardware_util.png` (CPU/GPU/ANE/W stacked timeline from `sudo powermetrics` over 60 s of decode) — lives in the same folder for appendix reference; the four embedded above are sufficient for the rubric defense.

4 · MSI Cut the Cord compliance + reproduction

Zero cloud AI at demo: MLX-LM in-process + Ollama embeddings + whisper.cpp + macOS `say`. `scripts/airplane-check.sh` greps every Python/TS/Rust source file for external URLs and exits 0. **100 % local:** `tcpdump -i any -nn` during a 90-s demo recorded 0 outbound packets; the active backend is published on `/ares/perf.llm_backend`. **Software integration:** Tauri opens NASA PDFs in macOS Preview at the cited chunk · native folder picker · macOS mic via Tauri capability · `say` system call. **Single machine:** all 4 personas + ASR + TTS + FAISS + Arrow tile cache + 3D renderer on the same M3 Pro. **Airplane test:** pass — demo loop survives Wi-Fi off identically.

Reproduce `bash scripts/setup.sh && bash scripts/download-mars-corpus.sh && . backend/.venv/bin/activate && pip install mlx mlx-lm pyarrow && python -c "from mlx_lm import load; load('mlx-community/Qwen2.5-3B-Instruct-4bit')` then `LLM_BACKEND=mlx npm run tauri dev`. Bench: `python benchmarks/houston/{cache_lattice_bench,throughput_bench --label mlx-qwen2.5-3b,agent_python_exec_bench,run_benchmarks}.py && python benchmarks/houston/make_figures.py`.

EdgeXpert forward path — `_build_generator(...)` is a one-line swap; tile cache + persona prefix + code-as-action sandbox are device-agnostic. Houston ports to the MSI EdgeXpert NPU without architectural rewrite once `mlx-lm` exposes a Core ML / NPU device target.